Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering fluorescent structures that overlap in projection requires measurements that distinguish their axial arrangements. We encode depth in the illumination of a fixed-focus fluorescence microscope by scanning a plane wave in angle through a fixed, defocused phase plate. We learn the plate through the optical forward model to reduce correlations between depth-dependent measurement signatures, without specimen examples or a reconstruction decoder. We compare the learned plate with an untrained random-phase plate on simulated measurements of 60 mouse pyramidal neuron crops, at eightfold measurement compression and matched nominal excitation dose. Mean axial-profile Wasserstein-1 (W1) error decreases from 0.37 *to* $0.22\ \mu\mathrm{m}$ *with total variation (TV), from* 0.58 *to* $0.29\ \mu\mathrm{m}$ *with pretrained plug-and-play (PnP), and from* 0.27 *to* $0.20\ \mu\mathrm{m}$ *with neuron-trained PnP. The gain extends to a feed-forward decoder trained separately for each plate. Controlled crossing-filament tests show improved axial separation under two of the three iterative methods. Improvements over the untrained plate persist with background and read noise, and at reduced dose. Our plate also gives lower error than the tested coherence- and Fisher-information-based designs. A plate trained end-to-end with a decoder on specimen examples differs from ours by less than* $0.01\ \mu\mathrm{m}$ *in mean error under the iterative methods; its more extensively trained decoder achieves* $0.03\ \mu\mathrm{m}$ *lower error. These results show that learning depth-discriminative illumination independently of a decoder improves axial recovery across the tested reconstruction methods.*

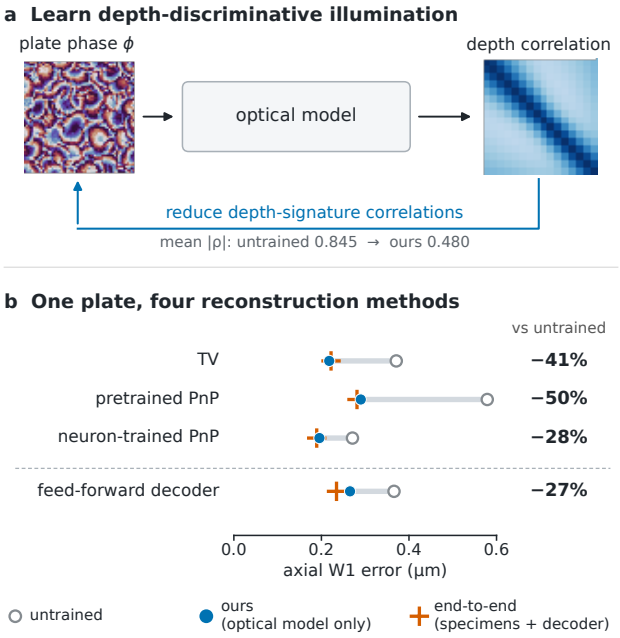


Figure 1. **Decoder-free optical learning improves axial recovery across four reconstruction methods.** (a) Training decorrelates detector-integrated depth signatures. The matrix shows our plate. Mean absolute off-diagonal correlations use 32 directions and 44×44 interior positions. (b) Mean W1 on simulated measurements at eightfold measurement compression ($C = 8$) and matched nominal dose. Percentages compare our plate with the untrained plate. Legend entries identify the inputs used for optical training. Feed-forward networks are fine-tuned for each plate, including the end-to-end plate. Paired intervals are in the supplement.

1. Introduction

Fluorescent branches that overlap in a camera image may lie at different depths. In fixed-focus fluorescence microscopy, a broad detection depth of field can make different axial arrangements produce similar images. Reconstruction then relies more heavily on prior assumptions [3, 11]. An accurate mean depth can still conceal two structures reconstructed as one. Faithful recovery therefore requires measurements that distinguish how fluorescence is distributed along depth.

We encode depth in the excitation by scanning the incident angle of a plane wave through a fixed, defocused phase plate (Fig. 2). Propagation produces structured illumination patterns that shift by different amounts at different depths as the angle changes. Structures at different axial positions therefore experience different sequences of excitation intensities. A camera records one fluorescence image per angle while the plate, specimen and detection focus remain stationary. A random phase plate already supports volumetric reconstruction, but the phase profile determines how clearly

the measurements distinguish axial structures.

We learn the phase profile to make the measurement responses of different depths more distinguishable. A voxel's measurement signature collects its camera responses across illumination angles, accounting for excitation, detection blur and pixel integration. We minimize weighted squared correlations between these signatures at different depths but the same lateral position (Fig. 1a). Training uses only the optical forward model, without specimen examples or a reconstruction network. The plate can therefore be designed before a reconstruction method is chosen.

Fisher information and worst-case coherence are established criteria for designing optics independently of a decoder [8, 26]. They emphasize sensitivity to small depth changes and the most similar pair of measurement responses, respectively. We compare our objective with these criteria under matched optimization budgets. We also evaluate end-to-end optical learning, which optimizes the phase plate jointly with a reconstruction network using specimen examples [5, 20].

We evaluate the optical designs on simulated measurements of mouse pyramidal neuron reference volumes from the public DEEP^2 dataset [29, 30]. Reconstruction uses total variation (TV), plug-and-play (PnP) with pretrained or neuron-trained denoisers, and a feed-forward decoder trained separately for each plate (Fig. 1b). Alongside intensity fidelity, we assess mean-depth placement and the full axial distribution of fluorescence. Neuronal examples illustrate the recovered profiles, while controlled filament crossings test the separation of overlapping structures.

Our contributions are threefold:

1. **Depth-specific optical learning.** We learn a defocused phase plate by minimizing weighted squared correlations between detector-integrated signatures of different depths at the same lateral position, without specimen examples or a reconstruction network.
2. **Competitive recovery across reconstruction methods.** At eightfold measurement compression and matched nominal excitation dose, our plate achieves 10–26 % lower mean axial-profile Wasserstein-1 error than Fisher-information and worst-case-coherence designs across four reconstruction methods. Under iterative reconstruction, mean error is close to that of the end-to-end plate.
3. **Improved axial structure recovery.** Relative to the untrained plate, axial-profile error decreases by 27–50 % across these methods. Controlled crossing tests also demonstrate improved filament separation under two of the three iterative methods.

2. Related work

Depth encoding in fluorescence microscopy. Depth can be encoded through detection optics or structured excitation. Light-field microscopy, DiffuserCam and Miniscope3D exploit depth-dependent detection responses [1, 4, 31]. On the excitation side, Pascucci *et al*. [21] recover volumes from lateral speckle scans through axial decorrelation, with saturation improving resolution, while Yeh *et al*. [32] combine translated speckles with axial scanning. Yilmaz *et al*. [33] use incident-angle scanning through a fixed scattering layer for two-dimensional fluorescence recovery; related work obtains depth cues from angular correlations [27] and tilted-pattern shifts [13]. These precedents motivate our random phase-plate baseline. We optimize the phase profile of a defocused plate for depth discrimination, keeping the plate fixed throughout acquisition.

Learning optical encoders. End-to-end approaches jointly optimize optics and reconstruction networks [12], including DeepSTORM3D [20] for emitter localization and FourierNets [6] for volumetric microscopy. Kellman *et al*. [16] optimize illumination for a specified reconstruction algorithm. Decoder-free design instead targets measurement properties. Shechtman *et al*. [26] use Fisher information to design point-spread functions (PSFs) for three-dimensional localization. Miniscope3D [31] aggregates cross-coherence between depth-dependent detection PSFs, using a smoothed maximum over lateral shifts for each depth pair. Squared-correlation objectives have a classical frame-potential interpretation [2], and related Gram-matrix criteria are established in sensing-matrix design [7]. IDEAL [23] and IDEAL-IO [19] optimize estimated mutual information using fitted measurement distributions, without a reconstruction decoder. Our objective specializes coherence-based design to detector-integrated voxel signatures across excitation angles, comparing depths at the same lateral position. Training uses the optical model alone, without specimen examples or a fitted measurement distribution.

Volumetric reconstruction and priors. Reconstruction methods range from total variation regularization [25] to plug-and-play denoisers [28] and learned networks following a model-based inverse [14]. Multi-slice fusion [17] combines lower-dimensional denoisers along complementary orientations of a shared volume. We use these complementary approaches to assess whether gains from learned illumination persist across different reconstruction priors.

3. Method

3.1. Depth-encoding acquisition

A plane wave illuminates a fixed phase plate with transmission $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagation converts its phase modulation into spatially structured excitation (Fig. 2a). Changing the illumination angle shifts the excitation pattern by an amount that varies with depth. In the ideal paraxial thin-screen approximation, a change $\Delta\mathbf{s}$ in transverse direction

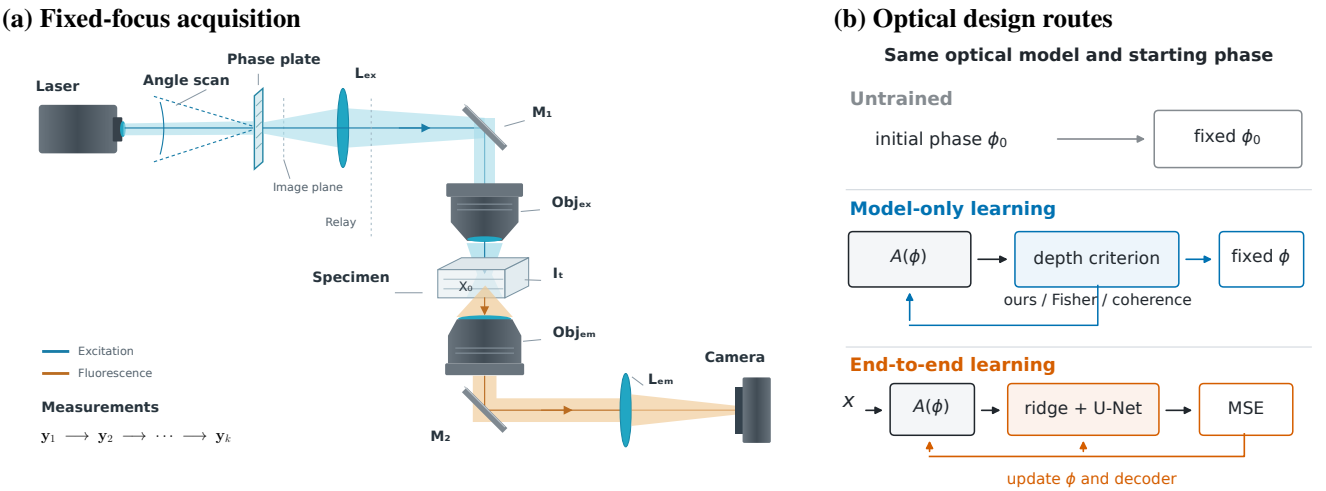


Figure 2. **A shared optical model supports three design routes.** (a) Only the incident angle changes during acquisition; the plate, specimen and detection focus remain stationary. Blue and orange indicate excitation and emission. (b) The untrained phase is retained, optimized using a model-only criterion, or optimized jointly with a decoder on specimen volumes. Each route yields a distinct fixed plate for reconstruction. Shared optics do not imply equal training budgets.

cosine produces a lateral displacement $\Delta\mathbf{r}(z) \approx d(z)\Delta\mathbf{s}$, where $d(z)$ is the axial distance from the plane conjugate to the plate [9, 10]. Plate defocus places this shear pivot outside the specimen, producing depth-dependent excitation sequences as the angle is scanned. We compute the actual excitation using a vectorial relay model that accounts for the high numerical aperture (NA), finite pupil and departures from pure translation [18, 24]. Acquisition is sequential: one fluorescence image is recorded per angle, while the plate, specimen and detection focus remain fixed.

Image formation and signatures. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation, with $\mathbf{r} = (x, y)$. Under linear, unsaturated fluorescence, illumination $I_t(\mathbf{r}, z; \phi)$ produces an emission source proportional to $X_0 I_t$. Contributions from different locations add incoherently through the detection optics. For a piecewise-constant reconstruction with coefficient x_j in voxel V_j, the noiseless model is

$$\bar{\mathbf{y}} = \mathsf{A}_\phi \mathbf{x},$$
$$(\mathsf{A}_\phi)_{(t,p),j} = \int_{V_j} I_t(\mathbf{r}, z; \phi)\, g_p(\mathbf{r}, z)\, \mathrm{d}^2\mathbf{r}\, \mathrm{d}z, \tag{1}$$

where $g_p(\mathbf{r}, z)$ is the response of camera pixel p to unit emission at $(\mathbf{r}, z)$, including the laterally shift-invariant emission point-spread function and pixel averaging [32]. Excitation modulates the specimen before detection blur, so the camera need not resolve the illumination pattern itself. We evaluate the voxel integrals on a finer optical grid to retain within-voxel variation; fine-grid measurement generation and specimen discretization error are detailed in the supplement. Each column of A_ϕ collects one voxel's responses across illumination angles and camera pixels, defining its measurement signature.

3.2. Phase-plate learning

Depths at the same lateral position have overlapping detection responses, so their excitation sequences provide an important source of axial discrimination. We optimize the phase plate to reduce correlations between the corresponding measurement signatures. For sampled illumination angles $\mathcal{T}$, let $\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z}$ denote the signature at lateral position $\mathbf{r}$ and depth z, centered over all angle–pixel entries and normalized to unit norm. The inner product $\rho_{zz'} = (\mathbf{a}^{\mathcal{T}}_{\mathbf{r},z})^\top \mathbf{a}^{\mathcal{T}}_{\mathbf{r},z'}$ measures correlation between two depths at that position.

We minimize the weighted mean squared correlation, averaged over lateral positions and sampled angle sets:

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}} \left[\frac{\sum_{z \neq z'} w_{zz'}\, \rho^2_{zz'}}{\sum_{z \neq z'} w_{zz'}} \right]. \tag{2}$$

The depth-pair weights $w_{zz'}$ are specified in the supplement. Squaring penalizes correlations of either sign. With uniform weights, the numerator is the off-diagonal energy of the depth-signature Gram matrix, equivalent to the classical frame potential up to a constant [2]. The supplement gives a statistical interpretation of the unweighted criterion.

Training uses the optical forward model alone, without specimen examples or a reconstruction solver. We optimize the phase map by backpropagating through excitation propagation, the vectorial relay and detector integration. Signature normalization prevents uniform power scaling from directly lowering the loss; nominal excitation dose is controlled separately during evaluation.

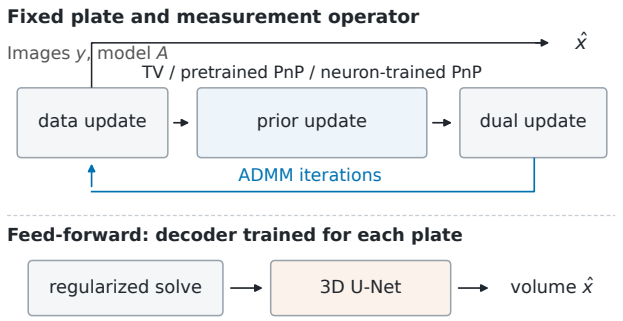


Figure 3. **Reconstruction is evaluated after the plate is fixed.** TV and PnP use the alternating direction method of multipliers (ADMM) with different prior updates. The feed-forward route uses a regularized solve followed by a 3D U-Net whose weights are trained separately for each plate. Full updates are in the supplement.

Alternative decoder-free objectives. We compare our objective with two established design criteria: worst-case coherence [8] and depth Fisher information [26]. The coherence objective targets the most strongly correlated depth pair at each lateral position through a smooth maximum of $|\rho_{zz'}|$. The Fisher objective minimizes $-\mathbb{E}_z \log \mathcal{I}_z$, where $\mathcal{I}_z$ measures local sensitivity to axial displacement under a Poisson model with unknown emitter brightness. Both are optimized using the same optical model and phase parameterization. Their full formulations and shared optimization settings are provided in the supplement.

End-to-end counterpart. Using the same optical model, we jointly optimize the plate and a decoder on neuronal training volumes (Fig. 2b). The decoder applies a regularized linear solve followed by a residual 3D U-Net. Training minimizes reconstruction mean squared error and updates both the phase map and decoder parameters. We then fix the learned plate and evaluate it with all reconstruction methods. The full decoder formulation and training protocol are provided in the supplement.

3.3. Reconstruction with fixed optics

We evaluate three iterative methods and a feed-forward decoder using the operator A_ϕ of each fixed plate (Fig. 3). TV combines a quadratic measurement term with isotropic 3D total variation and $[0, 1]$ intensity bounds [25]. PnP uses denoisers as implicit priors within ADMM [15, 28]. Following multi-slice fusion [17], pretrained PnP applies a natural-image DnCNN independently along three orthogonal slice orientations of a shared volume [34]. For slice denoising, neuron-trained PnP uses a 2D network trained on neuronal volumes.

The TV regularizer's form and PnP denoiser weights remain fixed across plates. For these iterative methods, scalar parameters are selected separately for each plate and acquisition condition using held-out validation volumes. Feed-forward decoders share an architecture, with weights fine-tuned separately for each fixed plate. Full solver updates, denoiser configurations and training protocols are provided in the supplement.

4. Experimental setup

Simulation settings. We simulate linear fluorescence in a homogeneous, non-scattering medium. Excitation and detection numerical apertures are 0.8 and 0.1, respectively. Both excitation and emission use a wavelength of $532\,\mathrm{nm}$ as a modeling approximation. The phase plate has $0.32\,\mu\mathrm{m}$ features and lies $6.6\,\mu\mathrm{m}$ before the intermediate illumination image plane, in specimen-equivalent coordinates. Optical fields are evaluated at $0.08\,\mu\mathrm{m}$ lateral and $0.32\,\mu\mathrm{m}$ axial spacing. Measurement generation uses an isotropic $0.32\,\mu\mathrm{m}$ analysis grid, and reference volumes are averaged to $0.64\,\mu\mathrm{m}$ reconstruction voxels.

Volumes and compression. Each reconstructed volume contains 16 axial planes of 48×48 voxels ($N = 36{,}864$), covering $30.72 \times 30.72\,\mu\mathrm{m}^2$ laterally and $10.24\,\mu\mathrm{m}$ axially. Each camera image contains 12×12 samples. With k illumination directions, measurement compression is $C = N/(144k)$. We evaluate $k = 256, 64$, and 32, corresponding to $C = 1, 4$, and 8, respectively. For $C > 1$, five randomly selected direction subsets are shared across plates.

Reference volumes. We construct neuronal reference volumes from public two-photon stacks of mouse pyramidal neurons in the DEEP^2 dataset [29, 30]. Crops are normalized and assigned the analysis-grid spacing defined above; reconstruction distances use these simulation coordinates. The test set contains 60 crops from six cells across three animals. Two crops per cell were prespecified using reference structure, and eight additional crops per cell were sampled randomly after reconstruction parameters were fixed. Other full-field studies use the 12 prespecified crops unless stated otherwise.

Training and validation. Neuron-trained denoisers and supervised models use 459 training crops from a fourth animal in the same DEEP^2 dataset. Supervised checkpoint selection uses 48 validation crops from a held-out stack of that animal. TV and PnP parameters are selected using two cortical vascular crops from a separate animal in the same public release. Test volumes are excluded from training, checkpoint selection and parameter tuning.

Compared plates. The primary comparison includes an untrained phase plate, plates optimized with our objective,

depth Fisher information and worst-case coherence, and an end-to-end trained plate. The untrained plate has independent phases drawn uniformly from $[0, 2\pi)$ at each feature and provides the common initialization for all learned plates. The three decoder-free objectives share the phase parameterization, sampled lateral positions and angles, optimizer settings and update budget. End-to-end training follows a separate supervised schedule with additional decoder updates. Full training protocols are provided in the supplement.

Excitation dose. We normalize each plate's excitation to the same mean over the specimen and full angular scan. Exposure per image increases in proportion to the reduction in image count, matching nominal excitation dose for the full scan and in expectation over direction subsets. For each specimen, exposure is calibrated so that the untrained plate yields 1000 expected photons per camera pixel per image at the full-scan mean signal. This calibration is shared across plates. Measurements include Poisson shot noise; detected photon counts can differ because each illumination pattern overlaps differently with the specimen.

Metrics and statistics. Our primary axial metric is one-dimensional Wasserstein-1 (W1) distance [22], which compares the full normalized intensity profile along depth. For each lateral column c, let F_c and $\hat{F}_c$ denote the cumulative distributions of the normalized reference and nonnegative reconstructed profiles. We compute

$$E_{\mathrm{W1}} = \sum_c \pi_c \, \Delta z \sum_{i=0}^{Z-2} \left| F_c(i) - \hat{F}_c(i) \right|, \qquad (3)$$

where $Z = 16$, $\Delta z = 0.64\,\mu\mathrm{m}$, and π_c is the reference column brightness normalized over columns exceeding 10 % of the maximum reference column brightness. Centroid error measures mean depth placement using the same columns and weights. PSNR and foreground SSIM complement these axial metrics by assessing intensity fidelity.

We compare plates on the same crops and illumination-direction subsets, at matched nominal dose and under the same noise model. For each crop, we average the reconstruction metrics over direction subsets and their associated noise realizations before calculating changes between plates. Crops from the same neuronal cell are related, so we estimate uncertainty by resampling cells. We construct 95 % confidence intervals using 10,000 bootstrap resamples of the six cells, retaining all crops belonging to each sampled cell.

5. Results

A random phase plate provides depth-dependent excitation during angular scanning. We use it as the baseline to measure what learning adds to the same optical system. Each learned plate starts from this phase map, with scan directions and nominal excitation dose kept the same. Three random phase realizations give similar centroid accuracy and image quality, supporting this baseline choice (supplement).

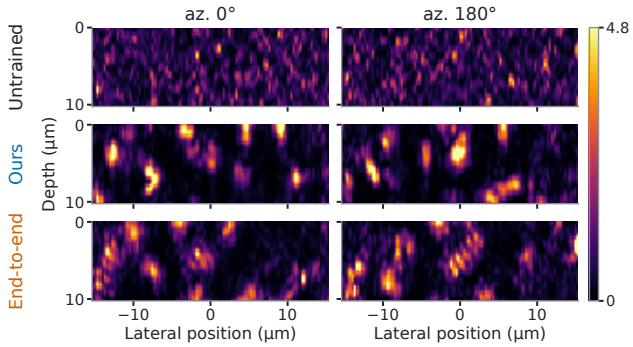


Figure 4. **Optical learning changes the depth code before reconstruction.** Illustrative excitation sections at equal and opposite tilts (azimuths 0° and 180°) in the same central vertical plane, for the untrained plate (top), ours (middle) and the end-to-end plate (bottom). The shared linear intensity scale is relative to the untrained mean.

Learning makes the defocused plate's depth code more selective (Fig. 4): strong responses occupy different axial regions as the incident angle changes. Over the interior field, mean absolute depth-signature correlation falls from 0.845 to 0.480 (Fig. 1a); the end-to-end plate reaches 0.628 (matrices in the supplement). These correlations use the full detector-integrated signatures, rather than illumination intensity alone.

5.1. One learned plate, four reconstruction methods

At $C = 8$ and matched nominal dose, our plate lowers mean axial-profile W1 by 41 % with TV, 50 % with pretrained PnP, 28 % with neuron-trained PnP and 27 % with the feed-forward decoder, relative to the untrained plate (Fig. 1b, Tab. 1). Paired 95 % cell-bootstrap intervals exclude zero for all four comparisons. W1 improves on all 60 crops under each TV/PnP method, covering both prespecified and later randomly sampled crops.

Centroid error also decreases by 42 %, 54 % and 30 % under TV, pretrained PnP and neuron-trained PnP, respectively (Tab. 1). PSNR increases by 1.4–1.8 dB across the four methods, accompanying the improvement in axial recovery. Our plate with TV also gives lower mean W1 than the untrained plate with neuron-trained PnP (0.218 versus $0.271\,\mu\mathrm{m}$).

5.2. Comparison with established optical objectives

All three decoder-free objectives reduce W1 relative to the untrained plate (Tab. 1). Under matched sampling and optimization budgets, our objective gives 12–26 % lower mean W1 than worst-case coherence and 10–21 % lower than depth

	Axial-profile W1 (μm)					Centroid error (μm)				
Reconstruction	Untrained	Depth Fisher	Worst-case coherence	Ours	End-to-end	Untrained	Depth Fisher	Worst-case coherence	Ours	End-to-end
TV	0.371	0.251	0.264	**0.218**	0.222	0.259	0.175	0.185	**0.150**	0.151
Pretrained PnP	0.579	0.367	0.390	0.290	**0.281**	0.490	0.298	0.320	0.224	**0.212**
Neuron-trained PnP	0.271	0.218	0.223	0.195	**0.189**	0.197	0.157	0.160	0.139	**0.131**
Feed-forward	0.366	0.314	0.314	0.265	**0.234**	0.282	0.234	0.238	0.196	**0.167**

Table 1. **Optical design evaluated across four reconstruction methods.** Mean axial-profile W1 and centroid error (μm) on simulated measurements at $C = 8$ and matched nominal dose: 60 crops from six cells and three animals, averaging five direction subsets per crop. Depth Fisher, worst-case coherence and ours are trained through the optical model only; end-to-end is trained jointly with a decoder. Model-only criteria use sampling seed 0 and matched optimization settings; bold and underline mark the lowest and second-lowest means for each metric. The feed-forward network is fine-tuned separately for each plate. All trained plates start from the untrained phase map. Paired intervals and intensity metrics are in the supplement.

Fisher information across the four reconstruction methods. Paired 95 % cell-bootstrap intervals exclude zero in each comparison. Centroid error also favors our plate over both alternatives under all four methods.

The TV/PnP advantage over both alternatives persists when optical training is repeated with a different sampling sequence from the same initial phase. On the 12 prespecified crops, omitting signature centering changes mean W1 from 0.206 to $0.207\,\mu$m with neuron-trained PnP.

5.3. Comparison with end-to-end training

Our plate and the end-to-end plate differ by less than $0.01\,\mu$m in mean W1 under TV and the two PnP methods (Tab. 1). The end-to-end plate has 1.9 % higher W1 under TV, but 3.0 % and 3.2 % lower W1 under pretrained and neuron-trained PnP, respectively. Paired 95 % cell-bootstrap intervals exclude zero for all three W1 comparisons. Centroid error favors end-to-end by 5.4 % under both PnP methods, with no established TV difference. The end-to-end plate also transfers beyond its training decoder.

With separately fine-tuned feed-forward decoders, the end-to-end system achieves lower W1 (0.234 versus $0.265\,\mu$m, a 12 % reduction) and 0.48 dB higher PSNR. Its decoder receives 15,500 updates in total, compared with 8000 for ours, so the comparison includes unequal decoder training histories. The TV/PnP comparison holds prior families and denoiser weights fixed, showing that competitive depth recovery can be achieved without specimen examples or decoder feedback during optical training.

5.4. Recovery of overlapping structures

Figure 5 shows a neuronal crop where both learned plates reduce crop-wide profile error. At the marked column, two unequal axial peaks share a lateral position: our plate retains the weaker component more faithfully, while the untrained and end-to-end reconstructions attenuate it. This local W1 advantage persists across all five direction subsets, although the two learned plates have similar crop-wide W1. Reference-selected views from two animals and all three iterative methods are provided in the supplement.

On reference columns with multiple axial peaks, our plate lowers W1 by 34 % with TV, 42 % with pretrained PnP and 28 % with neuron-trained PnP relative to the untrained plate. This descriptive comparison uses 59 eligible crops. Since multiple peaks can occur within a single branch, we use controlled filament crossings to test the separation of distinct structures.

At $C = 8$ and nominal dose, we evaluate three lateral crossing geometries and two axial centers, with five measurement realizations per case. Success requires two matched axial peaks in columns carrying at least half the reference crossing signal, together with correct depth ordering of the filament arms. The qualifying gap is the smallest tested separation achieving at least 80 % success at each center and at all larger tested gaps. Confidence intervals resample phantom cases after averaging their measurement realizations.

Relative to the untrained plate, our plate reduces the qualifying gap from 3.84 to $2.56\,\mu$m with TV and pretrained PnP (Fig. 6). The gap remains $3.84\,\mu$m under neuron-trained PnP despite improved neuronal W1. The zero-gap control shows no false splitting. At nominal dose, paired intervals for differences in pooled crossing success between our plate and the end-to-end plate include zero under all three methods.

At one tenth of nominal dose, we pool crossing success over gaps of 1.28, 1.92 and $2.56\,\mu$m, using three measurement realizations per case. Our plate achieves 17 and 15 percentage points higher success than the end-to-end plate under TV and pretrained PnP, respectively (Fig. 6c). Both plates have low success under neuron-trained PnP, with no established difference. The supplement provides full gap curves and tests with weaker or additional filaments.

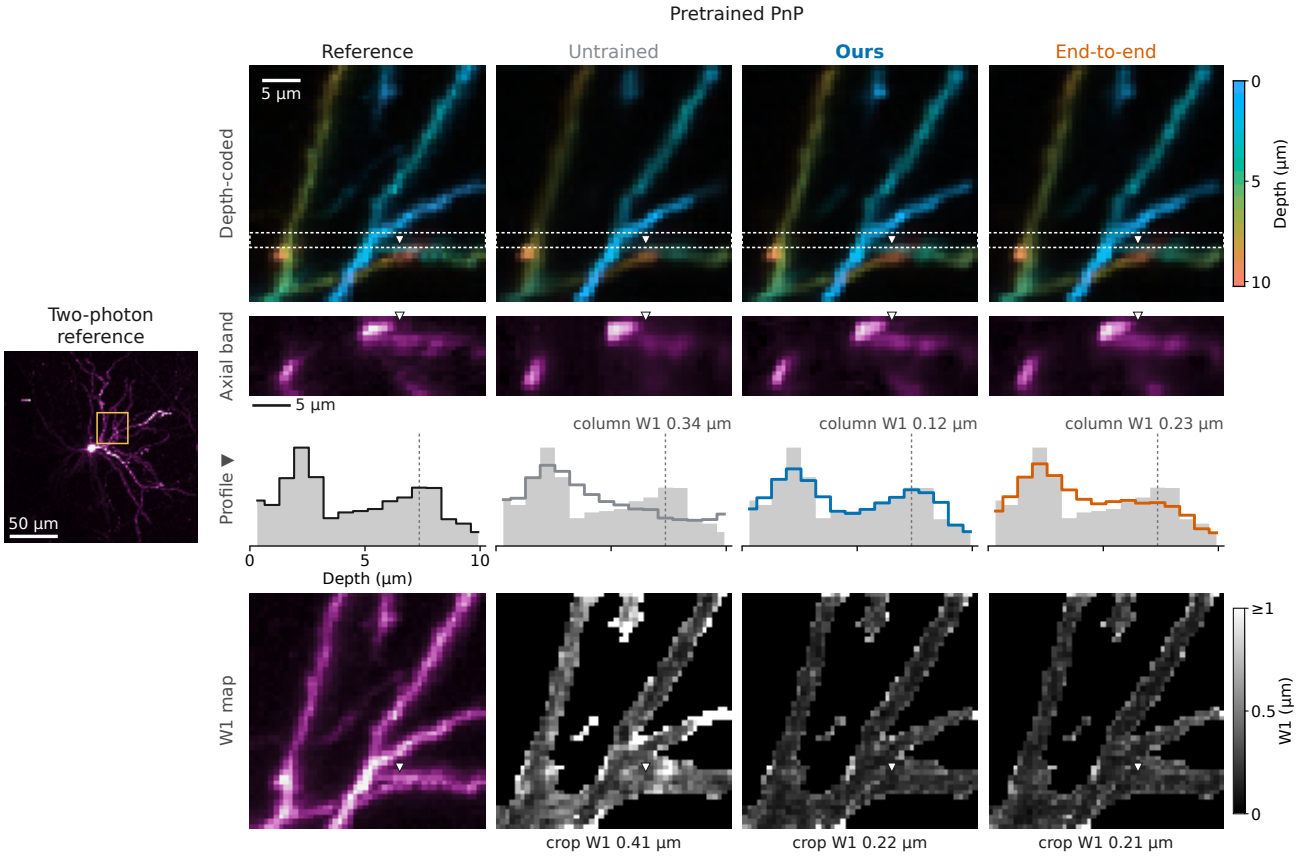


Figure 5. **Recovery of unequal axial peaks.** Matched reconstructions with pretrained PnP at $C = 8$. The marked column, chosen after inspecting reconstructions, illustrates better retention of the weaker peak by our plate (dashed depth line). Profiles have unit mass, with the reference in gray; column and crop W1 are labeled separately. The bottom-left panel shows reference intensity. Fluorescence images use magenta; W1 maps run from black (0) to white ($\geq 1\,\mu$m), with black also marking pixels outside the reference signal mask. Reconstruction images share a linear intensity scale and noise draw. Reconstructed views use simulation coordinates; the two-photon context uses tissue coordinates. Five-subset profiles and selection details are in the supplement.

5.5. Acquisition budgets and calibration

On the 12 prespecified crops, our plate with 32 images gives lower mean W1 than the untrained plate with 256 images under TV (0.196 versus 0.249 μm) and pretrained PnP (0.277 versus 0.413 μm). Neuron-trained PnP gives close means (0.206 versus 0.203 μm). These comparisons use matched nominal dose, with longer exposure per image for the shorter scan.

On the same 12 crops at $C = 8$, our plate at 0.3 times nominal dose gives 22–39 % lower W1 than the untrained plate at nominal dose across TV and the two PnP methods. At matched nominal dose, the gain also persists with added background and read noise. The supplement provides the full dose and noise comparisons.

With 256 images at nominal dose, our plate has lower W1 than the end-to-end plate under pretrained PnP. At $C = 8$ and one tenth of nominal dose, the end-to-end plate gives lower neuronal W1 under neuron-trained PnP (0.227 versus 0.239 μm). In each setting, differences under the other two iterative methods are not established.

At $C = 8$ and nominal dose, our plate retains 27–28 % lower W1 than the untrained plate for known plate-distance offsets of $\pm 0.5\,\mu$m, using neuron-trained PnP. The operator is updated and scalar reconstruction parameters are revalidated, without retraining the plate. Ignoring distance or lateral offsets can reverse the advantage: a 0.32 μm lateral shift makes our plate's W1 11 % higher than the untrained control under the same shift (95 % CI: 2–23 %).

6. Discussion and limitations

Decoder-free optical design. Our results support depth-signature decorrelation as a useful criterion for learning illumination independently of reconstruction. Under matched optimization budgets, our objective improves axial recov-

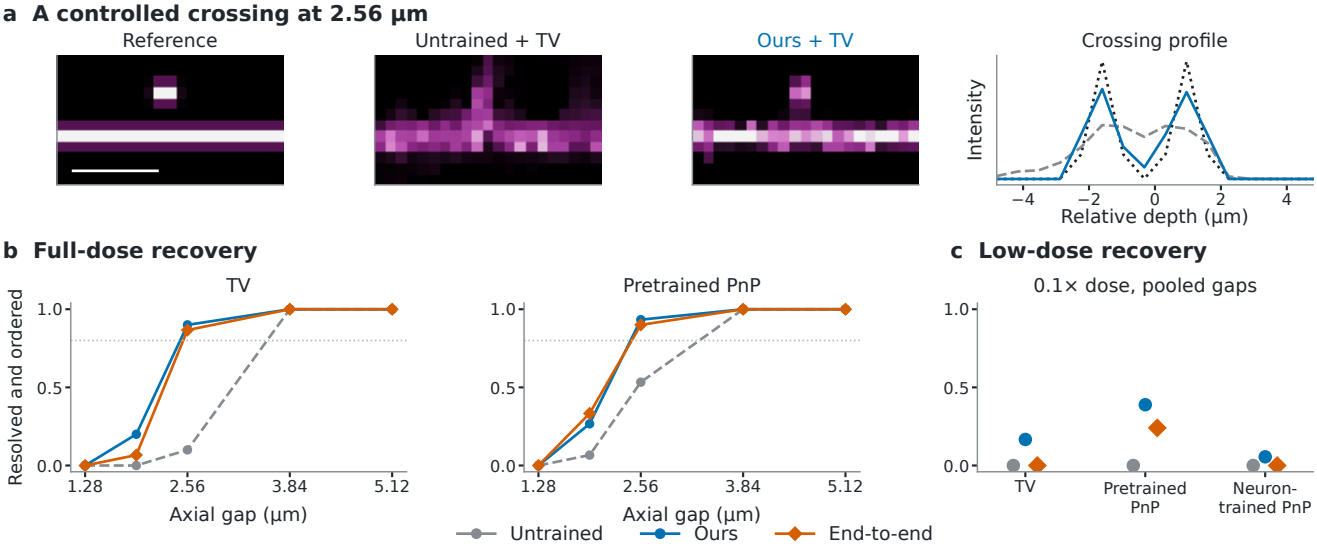


Figure 6. **Controlled structure recovery depends on the code, prior and dose.** (a) Matched fluorescence sections (magenta) and profiles through one crossing at our plate's TV threshold gap; the first fixed geometry, center and draw are shown, with a shared intensity scale. Bar: 5 μm. (b) Full-dose success rates pool the two axial centers; the 80 % criterion is applied to each center separately. (c) Reduced-dose success pooled over all three positive gaps (1.28–2.56 μm), for each iterative method. Full gap curves and paired phantom-bootstrap intervals are in the supplement.

ery over the tested coherence and Fisher-information criteria across four reconstruction methods. End-to-end optical learning also transfers to iterative methods and achieves lower error with its own decoder, despite producing more correlated depth signatures. Lower correlation therefore does not guarantee better reconstruction. The significance of our approach is that it improves depth encoding without specimen examples or reconstruction feedback, leaving decoder choice outside optical training.

Axial recovery and structure. Lower axial-profile error indicates a closer match to the reference depth distribution, but does not by itself establish that overlapping structures are resolved. Controlled crossings provide a direct test: our plate separates filaments at smaller gaps than the untrained plate under TV and pretrained PnP, while neuron-trained PnP retains the same qualifying gap despite improved neuronal W1. The structural benefit of learned illumination therefore depends on the reconstruction prior and acquisition conditions, as the reduced-dose comparisons further illustrate.

Physical implementation. Our simulations assume linear fluorescence in a homogeneous medium with an ideal thin phase plate, and use the same optical model for measurement generation and reconstruction. The learned patterns concentrate excitation more strongly, so operating power must keep fluorescence unsaturated. A physical implementation requires characterization of the realized excitation and detection responses. The alignment simulations show that updating the operator for known offsets can preserve the learned advantage, whereas unmodeled misalignment can reverse it.

Scope of the evidence. Our evaluation uses six cells from three animals, with dependent crops and approximate cell-bootstrap intervals. Optical learning is evaluated from one initial phase map, and the end-to-end comparison uses one architecture and training run. Supervised training uses coarse-basis measurements with Gaussian noise, whereas evaluation retains fine-grid specimen variation and Poisson counts. The relative performance of the designs should therefore be interpreted within these sampling and training conditions.

7. Conclusion

We learn depth-encoding illumination from the optical model alone, without specimen examples or a reconstruction decoder. In fixed-focus fluorescence simulations, the same plate lowers mean axial-profile W1 from 0.27–0.58 μm to 0.20–0.29 μm across four reconstruction methods at eightfold measurement compression. It also separates crossing filaments at smaller gaps under two of the three iterative methods. Mean W1 differs from the end-to-end plate by less than 0.01 μm under the iterative methods; the end-to-end system gives lower error with its own decoder. Depth signatures can therefore guide optical learning while leaving reconstruction choice outside the training loop.

LLM Usage Statement

OpenAI ChatGPT/Codex and Anthropic Claude assisted with experimental planning and code, manuscript and supplement organization, drafting and editing, literature checks, numerical audits of simulation outputs, and figure/table preparation. The authors retain responsibility for the methods, results, citations and final manuscript.

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